

1st Year Exam: Numerical Analysis, Jan, 2025.

Do 4 (four) problems.

1.  $g(x) = \frac{x}{2} + \frac{a}{2x}$  has  $x = \sqrt{a}$  as a fixed point where  $a > 0$ . Suppose we use the fixed point method with a starting point  $x_0 > \sqrt{a}$ . Use the theory of the fixed point method to determine the condition on  $x_0$  which guarantees

$$|x_{k+1} - \sqrt{a}| \leq \frac{1}{2}|x_k - \sqrt{a}|, \quad k \geq 1.$$

2. Let  $f \in C[0, 1]$  and let  $\sum_{i=1}^n w_i f(x_i)$  ( $n \geq 1$ ) be an approximation to  $\int_0^1 f(x) dx$ . Assume that  $0 \leq w_i \leq 1, 0 \leq x_i \leq 1$  ( $i = 1, 2, \dots, n$ ). Let

$$E_n(f) = \int_0^1 f(x) dx - \sum_{i=1}^n w_i f(x_i)$$

be the error in the approximation. Assume that  $E_n(p_n) = 0$  for  $p_n$ , which denotes any polynomial of degree  $\leq n$ . Use the Weierstrass approximation theorem to prove that, given  $\epsilon > 0$ , there is an  $N > 0$  such that  $|E_n(f)| < \epsilon$  when  $n > N$ .

3. Given a differentiable function  $f(x)$ , consider the problem of finding a polynomial  $p(x) \in \mathbb{P}^n$  such that

$$p(x_0) = f(x_0), \quad p'(x_i) = f'(x_i), \quad i = 1, 2, \dots, n,$$

where  $x_i, i = 1, 2, \dots, n$ , are distinct nodes. (It is not excluded that  $x_1 = x_0$ .) Show that the problem has a unique solution and describe how it can be obtained.

4. Let  $\{p_i(x)\}_{i=0}^\infty$  be the sequence of orthogonal polynomials obtained from the Gram-Schmidt orthogonalization process of  $\{x^i\}_{i=0}^\infty$ . Consider the quadrature formula  $Q(f) = \sum_{j=0}^m \alpha_j f(x_j)$  for  $J(f) = \int_a^b f(x) \rho(x) dx$ . Prove that  $Q(f)$  is exact for polynomials of degree  $\leq 2m + 1$  if and only if the quadrature nodes  $x_j$  ( $j = 0, 1, \dots, m$ ) are the roots of  $p_{m+1}(x)$  and the quadrature weights  $\alpha_j = \int_a^b l_j(x) \rho(x) dx, (j = 0, 1, \dots, m)$ .

5. Consider numerically solving the initial value problem

$$y'(t) = f(t, y), \quad 0 < t \leq t_f, \quad \text{with } y(0) = \eta.$$

Assume  $f$  is sufficiently differentiable and let  $h$  denote the step size. Show that the method

$$y_{n+2} - y_{n+1} = \frac{1}{4}h(f_{n+2} + 2f_{n+1} + f_n)$$

is  $A_0$  stable.